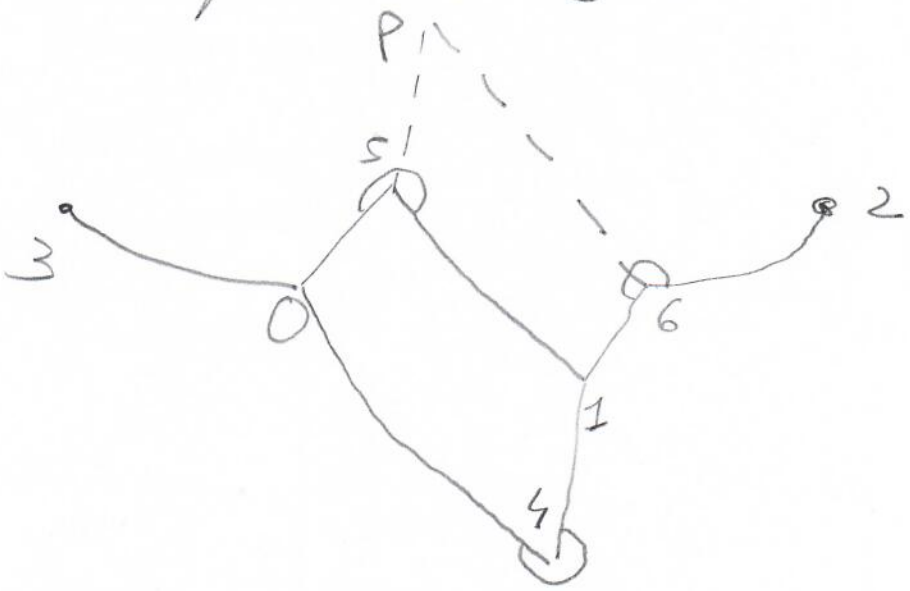
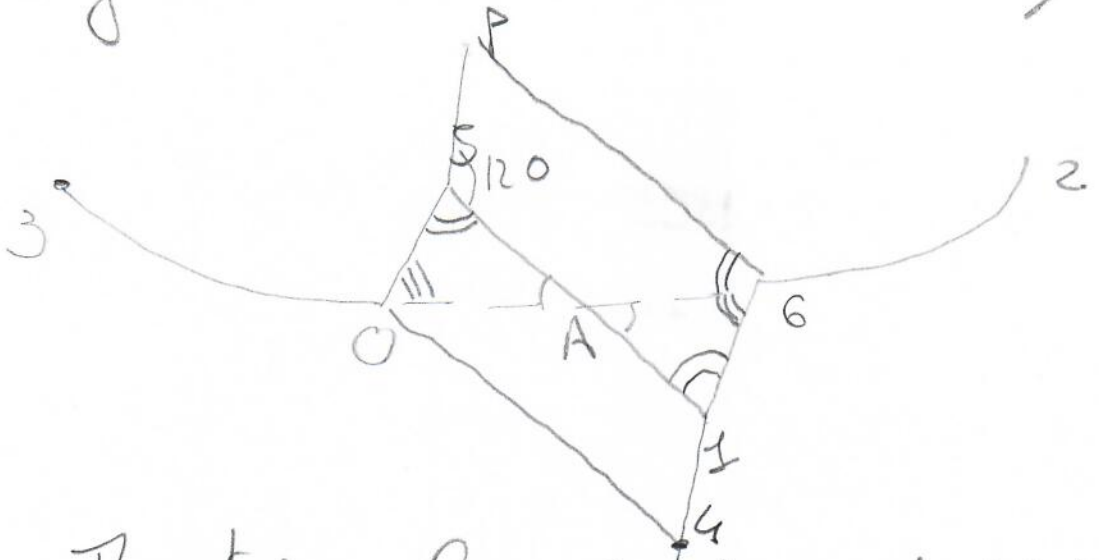
Escultura rete 6150

Observe that, because 2 and 3 are the vertices of a biangle, the vertices 3, 0, 6 and 2 are on the same equator γ. We draw a picture of the net in the half sphere which excludes the biangle and is cut in two equal quarters by γ



We also draw the location of the pin P.

Join O and 6 with the geodesic segment which is on the equator &



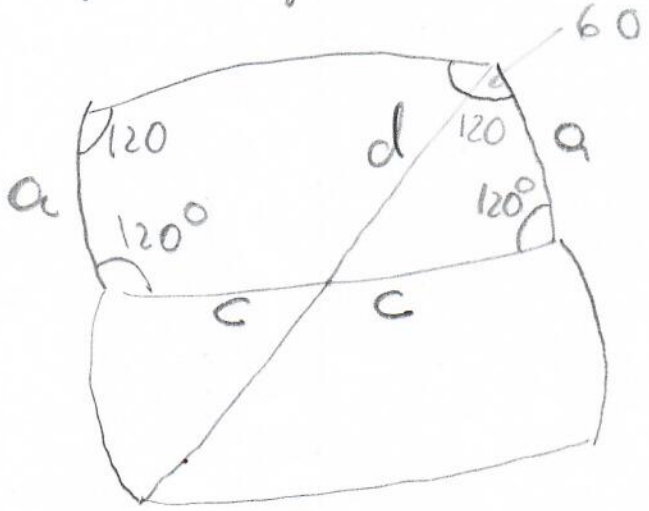
The triangles O5A and 16A have three equal angles and are therefore congruent.

So looking at the two quadrilaterals P516 and 4015 we have:

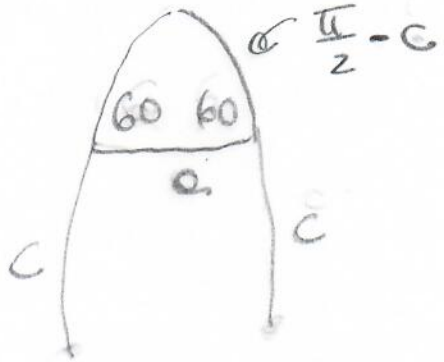
a) the edge in 51 in common
b) the edges 61 and O5 congruent
c) the angles in 1 and 5 equal
d) the angles in 6 and O equal

The two quadrilaterals are thus congruent.

Moreover they are determined by the fact that the bisector of one angle cuts the edge in 2 equal parts

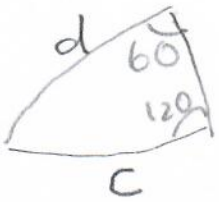


Now let us use Todhunter formulas



$$\tan\left(\frac{\pi}{2}-c\right)=\frac{\cos\theta}{\cos 60}\tan\frac{a}{2}$$

$$\tan\left(\frac{\pi}{2}-c\right)=2\tan\frac{a}{2} \quad (1)$$



$$\tan\frac{d+c}{2} = \frac{\cos 30}{\cos 90}\tan\frac{a}{2}$$

$$\Rightarrow \frac{d+c}{2} = \frac{\pi}{2}$$

$$\tan\frac{c-d}{2} = \frac{\sin 30}{\sin 90}\tan\frac{a}{2}$$

$$= \frac{1}{2}\tan\frac{a}{2}$$

$$\tan\left(c-\frac{\pi}{2}\right) = \frac{1}{2}\tan\frac{a}{2} \qquad (2)$$

(1) and (2) are not compatible